

Laboratory Audit of (lab name and lab identifier):**ISO/IEC 17025 Cert. identifier:** [\[see OCM opening / closing conference form\]](#)**Audit/Inspection Date(s):** [\[see OCM opening / closing conference form\]](#)**OCM Auditor(s)/Inspectors:** [\[see OCM opening / closing conference form\]](#)**Calibration and Maintenance of Laboratory Equipment Standards**

ID	Citation	Does the laboratory comply with this section?	Yes	No	N/A	Notes
1	OCM Cannabis Laboratory Quality System Standard, Section V	A laboratory must calibrate and maintain its equipment as specified below, and the calibration, verification and/or check and maintenance must be documented.				See Part 130.26 (a) (vii).
Air in a microbiological lab						
2	LQSS V, 1) a	Monitor bacterial density monthly				
Autoclaves						
3	LQSS V, 2) a	Check temperature with max-registering device weekly				
4	LQSS V, 2) b	Check performance with bioindicator monthly				
5	LQSS V, 2) c	Check timing quarterly				
6	LQSS V, 2) d	Use heat-indicating tape with each cycle				
Automatic Pipettes or Micropipettors and Dispensettes, and Pipette Tips						
7	LQSS V, 3) a	Check dispensing accuracy and precision quarterly				
8	LQSS V, 3) b	Calibrate annually				
9	LQSS V, 3) c	tips				
10	LQSS V, 3) d	Discard pipette tips after each use				
Balances						
11	LQSS V, 4) a	Check zero daily before use				
12	LQSS V, 4) b	Check accuracy with at least 2 weights over the range of use daily before use				
13	LQSS V, 4) c	Service and recalibrate annually				
14	LQSS V, 4) d	Keep service record				
Biosafety Cabinet - Class 1 or 2						
15	LQSS V, 5) a	Inspect for airflow with each use				
16	LQSS V, 5) b	Have certified annually				
Blenders (Autoclavable)						
17	LQSS V, 6) a	Clean and autoclave after each homogenization of submitted cannabis or cannabis products to be composited per lot				
18	LQSS V, 6) b	Alternatively, use other equipment to homogenize the cannabis or cannabis products to be composited per lot and ensure sterility is maintained between each lot.				
Cell Spreaders and Loops						
19	LQSS V, 7) a	Check sterility with each lot, or alternatively, keep certificate of conformance with each lot				
Centrifuges						
20	LQSS V, 8) a	Follow manufacturer's instructions.				

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Conductivity meter						
21	LQSS V, 9) a	Calibrate monthly				
Dilution water bottles						
22	LQSS V, 10) a	Check sterility, pH, and volume with each batch or lot				
Freezers						
23	LQSS V, 11) a	Check temperature daily				
24	LQSS V, 11) b	Defrost annually, if not self defrosting				
25	LQSS V, 11) c	Use temperature devices with temperature increments of 1 or 0.5°C				
Glassware for Chemistry Laboratory						
26	LQSS V, 12) a	Inspect for cleanliness, chips, and etching with each use				
27	LQSS V, 12) b	Use class A when specified by the approved method, and keep certificate of conformance per each piece of class A glassware				
28	LQSS V, 12) c	If class B or class A without a certificate of conformance, perform verification check upon purchase or prior to first use				
Glassware for Microbiological Laboratory						
29	LQSS V, 13) a	Inspect for cleanliness, chips, and etching with each use				
30	LQSS V, 13) b	See 12 b and c above.				
31	LQSS V, 13) c	Check pH with bromothymol blue with each wash batch				
32	LQSS V, 13) d	Conduct inhibitory residue test with initial use and new washing procedure (also may be annual)				
33	LQSS V, 13) e	Check for autofluorescence if used for testing with each batch or lot				
Hot-block digestors						
34	LQSS V, 14) a	Perform temperature distribution and stability study initially upon purchase and after any major maintenance / repairs				
Hot-air sterilizing oven for microbiological lab						
35	LQSS V, 15) a	Check temperature with each use				
36	LQSS V, 15) b	Check performance with bioindicator monthly				

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Incubators						
37	LQSS V, 16) a	Perform temperature distribution and stability study initially upon purchase and after any major maintenance / repairs				
38	LQSS V, 16) b	Check temperature daily if using continuous temperature monitoring device with data logger and capability to notify analyst or technician of any deviations, or twice daily for other temperature monitoring devices				
39	LQSS V, 16) c	Use temperature monitoring devices with temperature increments of 0.5°C or less, as appropriate (e.g., for a 44.5 +/- 0.2°C water bath used for incubation of thermotolerant bacteria, use a thermometer with 0.1°C increments)				
Laminar Flow Hoods						
40	LQSS V, 17) a	Check and service annually				
41	LQSS V, 17) b	Keep service record				
Media and reagent dispensing apparatus						
42	LQSS V, 18) a	Check volume dispense accuracy with each volume change				
43	LQSS V, 18) b	Check apparatus' accuracy by dispensing a sample volume of medium into a graduated cylinder just after filling/refilling it and periodically throughout extended runs; record results.				
44	LQSS V, 18) c	Lubricate parts according to manufacturer's instructions or at least once per month.				
Microbiological membrane filters and filter apparatus						
45	LQSS V, 19) a	Check sterility and properties with each new lot				
46	LQSS V, 19) b	Check sterility pre and post filtration series				
Microscopes						
47	LQSS V, 20) a	Clean optics and stage				
48	LQSS V, 20) b	Check alignment with each use				
49	LQSS V, 20) c	Follow manufacturer's instructions				
50	LQSS V, 20) d	Keep any service records				
Microwave Digestors						
51	LQSS V, 21) a	Follow manufacturer's instructions				
52	LQSS V, 21) b	Keep any service records				

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Muffle Furnace						
53	LQSS V, 22) a	Check accuracy annually using one of the following: 1) high-temperature thermometer with range to at least 500 C and with readable subdivisions of 5 C or less; 2) melting-point solids with capability of differentiating 5 C differences between 400 C and 500 C; or 3) independent potentiometer capable of differentiating 5 C differences between 400 C and 500 C				
54	LQSS V, 22) b	Keep any service records				
pH meters						
55	LQSS V, 23) a	Standardize with at least 2 buffer solutions daily before use				
56	LQSS V, 23) b	Determine slope daily before use				
Plastic or other non-glass volumetrics						
57	LQSS V, 24) a	Prior to use, keep certification of conformance per lot				
Refrigerators						
58	LQSS V, 25) a	Check temperature daily				
59	LQSS V, 25) b	Use temperature devices with temperature increments of 1 or 0.5°C				
Spectrophotometers						
60	LQSS V, 26) a	Check wavelength annually				
61	LQSS V, 26) b	Keep any service records				
Syringes						
62	LQSS V, 27) a	Keep certificate of conformance per syringe				
63	LQSS V, 27) b	Clean after use				
Temperature devices						
64	LQSS V, 28) a	Working units: For most types, check accuracy annually and establish a correction factor				
65	LQSS V, 28) Table 2	Liquid spirit in glass - frequency = annually				
66	LQSS V, 28) Table 2	Hg in glass- frequency = annually				
67	LQSS V, 28) Table 2	Digital, thermocouple or similar device- frequency = annually				
68	LQSS V, 28) Table 2	Dial- frequency = quarterly				
69	LQSS V, 28) Table 2	IR gun* frequency = quarterly (* In addition, check at single point daily on day of use)				
70	LQSS V, 28) b	Reference units: Recertify every 5 years				

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Timers						
71	LQSS V, 29) a	Autoclave: Check timing with stopwatch quarterly				
72	LQSS V, 29) b	Stopwatch: Check against National Time Signal (i.e., cell phone) annually				
UV-lamps and short-wave disinfection						
73	LQSS V, 30) a	Monitor bulb use with each use				
74	LQSS V, 30) b	Test with UV meter or perform plate count check quarterly				
Water baths for microbiological labs						
75	LQSS V, 31) a	Check temperature daily if using continuous temperature monitoring device with data logger and capability to notify analyst or technician of any deviations, or twice daily for other temperature monitoring devices				
76	LQSS V, 31) b	Use temperature monitoring devices with temperature increments of 0.5°C or less, as appropriate (e.g., for a 44.5 +/- 0.2°C water bath used for incubation of thermotolerant bacteria, use a thermometer with 0.1°C increments)				
77	LQSS V, 31) c	Fill unit only with reagent-quality water. Maintain water level so it is above the upper level of the medium in either tubes or flasks				
78	LQSS V, 31) d	Equip water bath with a gable cover to prevent evaporation and with a circulating pump to maintain even temperature distribution. Use only stainless steel, plastic-coated, or other corrosion-proof racks. Use screens or weights to keep materials from floating				
79	LQSS V, 31) e	Empty and clean bath as needed to prevent buildup of salts and microbial growth, and disinfect before refilling				
Weights						
80	LQSS V, 32) a	Working weights: Check with reference weights annually				
81	LQSS V, 32) b	Reference weights: Recertify every 5 years or more often (annually) if heavily used				
82	LQSS V, 32) c	Keep recertification record				

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Analytical Instrumentation						
83	Water Activity Meter: LQSS V, 33) a	Follow manufacturer's instructions for cleaning and maintenance				
84	Water Activity Meter: LQSS V, 33) b	Document all cleaning, calibrations, maintenance, and repairs				
85	Water Activity Meter: LQSS V, 33) c	Keep any service records				
86	Moisture Analyzer: LQSS V, 33) a	Follow manufacturer's instructions for cleaning and maintenance				
87	Moisture Analyzer: LQSS V, 33) b	Document all cleaning, calibrations, maintenance, and repairs				
88	Moisture Analyzer: LQSS V, 33) c	Keep any service records				
89	HPLC: LQSS V, 33) a	Follow manufacturer's instructions for cleaning and maintenance				
90	HPLC: LQSS V, 33) b	Document all cleaning, calibrations, maintenance, and repairs				
91	HPLC: LQSS V, 33) c	Keep any service records				
92	LC-MS: LQSS V, 33) a	Follow manufacturer's instructions for cleaning and maintenance				
93	LC-MS: LQSS V, 33) b	Document all cleaning, calibrations, maintenance, and repairs				
94	LC-MS: LQSS V, 33) c	Keep any service records				
95	GC-MS: LQSS V, 33) a	Follow manufacturer's instructions for cleaning and maintenance				
96	GC-MS: LQSS V, 33) b	Document all cleaning, calibrations, maintenance, and repairs				
97	GC-MS: LQSS V, 33) c	Keep any service records				

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Analytical Instrumentation						
98	ICP-MS: LQSS V, 33) a	Follow manufacturer's instructions for cleaning and maintenance				
99	ICP-MS: LQSS V, 33) b	Document all cleaning, calibrations, maintenance, and repairs				
100	ICP-MS: LQSS V, 33) c	Keep any service records				
101	Automatic plate readers and/or TEMPO: LQSS V, 33) a	Follow manufacturer's instructions for cleaning and maintenance				
102	Automatic plate readers and/or TEMPO: LQSS V, 33) b	Document all cleaning, calibrations, maintenance, and repairs				
103	Automatic plate readers and/or TEMPO: LQSS V, 33) c	Keep any service records				
104	Thermocyclers/PCR: LQSS V, 33) a	Follow manufacturer's instructions for cleaning and maintenance				
105	Thermocyclers/PCR: LQSS V, 33) b	Document all cleaning, calibrations, maintenance, and repairs				
106	Thermocyclers/PCR: LQSS V, 33) c	Keep any service records				